

RECITATION 1

PROBABILITY REVIEW, DECISION TREES

10-701: INTRODUCTION TO MACHINE LEARNING

September 4, 2026

1 Probability Review

1.1 Joints and Marginals

from 10-301/601 S26

Let A take values a_1, a_2 and let B take values b_1, b_2 , with $P(A = a_1) = 0.5$ and $P(B = b_1) = 0.5$.

1. Suppose a_1 and b_2 are disjoint (mutually exclusive). Fill in the full joint distribution $P(A, B)$, and give $p(a_1, b_1)$ and $p(a_1 \mid b_2)$.
2. Suppose instead that A and B are independent. Fill in the same table and give the same two quantities.
3. The two tables have identical marginals. What does that tell you about recovering a joint distribution from its marginals?

1.2 Distributions and Densities

1. Let $X \sim \text{Uniform}(0, \frac{1}{10})$, with density p_X . What is $p_X(0.05)$? What is $P(X = 0.05)$? Reconcile the two answers.
2. Is $p(x) = \frac{1}{2\sqrt{x}}$ on $(0, 1]$ a valid probability density?
3. Let $X \sim \text{Uniform}(0, 1)$ and let $Y = X^2$. Derive the density p_Y and compare it to the previous part.

1.3 Conditional Distributions and Factorization

Consider the conditional distribution $P(X, Y \mid Z)$ below, and suppose $P(Z = 0) = 0.4$.

$P(X, Y \mid Z)$ $X, Y \downarrow$	$Z \rightarrow$	0	1
00		0.1	0.5
01		0.2	0.3
10		0.6	0.2
11		0.1	0.0

1. Compute the joint entries $p(X = 1, Y = 0, Z = 0)$ and $p(X = 1, Y = 0, Z = 1)$. Which factorization did you use?
2. Compute $P(Z = 0 \mid X = 1, Y = 0)$.
3. Sum the row $(X, Y) = (1, 0)$ across the two columns. Why is it not 1, and what *is* guaranteed to sum to 1 in this table?

1.4 Expectation

1. **Tail-sum formula.** Let X be a nonnegative integer-valued random variable. Show that

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} P(X \geq k).$$

credit: 10-406/605 F25

2. **A distribution that is neither discrete nor continuous.** Let $Z \sim \mathcal{N}(0, 1)$ and let $X = \max(0, Z)$, a ReLU applied to a Gaussian pre-activation. Compute $P(X = 0)$; can X have a density in the usual sense? Compute $\mathbb{E}[X]$.

1.5 Bayes' Rule

You are training a binary classifier to detect spam emails! We know that only 1 in 1000 emails are actually spam. You train a classifier that outputs positive label with:

- [illegible]

2 Decision Trees

2.1 Entropy

Recall from lecture that the entropy of a distribution for a discrete random variable X is defined as

$$H(X) = \sum_c -P(X = c) \log_2 P(X = c)$$

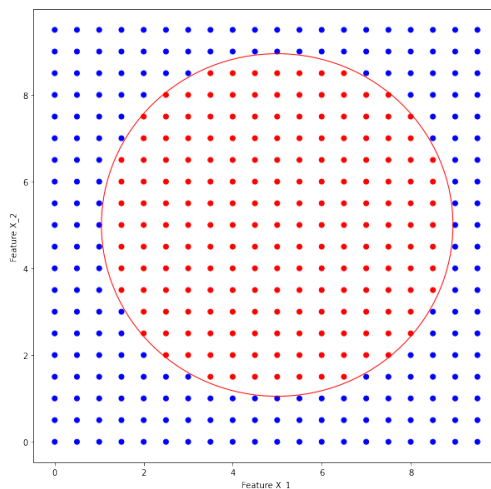
Let us define a random variable $X \in \{0, K - 1\}$. Define $\alpha_k = P(X = k)$. Show that the uniform distribution maximizes the entropy, that is:

$$\max_P H(X) = P(X = k) = \begin{cases} \frac{1}{K} & \text{if } k \in \{0, K - 1\} \\ 0 & \text{else} \end{cases}$$

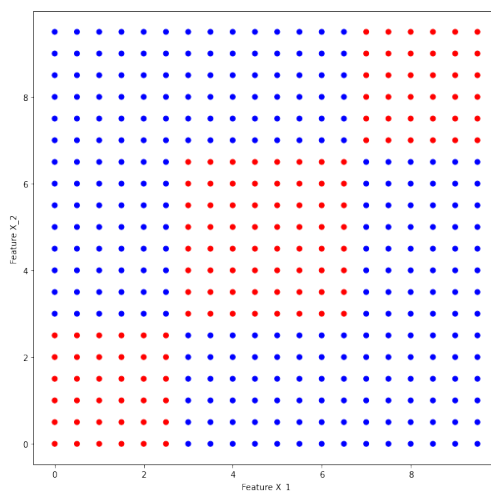
HINT: You will want to introduce a Lagrange multiplier constraint to enforce that the probability terms sum to 1. This constraint is of the form $\sum_k \alpha_k = 1$.

2.2 Decision Tree Decision Boundaries

1. Can a learned, finite decision tree using the ID3 algorithm perfectly classify all of the points in the figure below? Can it perfectly learn the decision boundary (portrayed as a circle in red)?



2. Consider the dataset below with 400 total points consisting of three clusters of red points with 36, 64, and 36 points going from bottom left to top right. The mutual information represents how much we gain, in terms of reducing the entropy, from knowing something about an attribute: $I(Y; X) = H(Y) - H(Y|X)$, where $H(Y|X) = \sum_k p(X = k)H(Y|X = k)$ represents the conditional entropy. Starting with the base predictor, what is the information gain (or mutual information) of the split $X_1 < 3$?



3. What is the mutual information of the split $X_2 < 3$ (we are performing this split AFTER the previous split, i.e. on the points with $X_1 < 3$).

4. Draw out a decision tree that could perfectly classify the points in the figure from Question 2. Why can we exactly learn the decision boundary in this case?

2.3 Bonus: KL Divergence

A concept that may be useful for your homework is the KL divergence, a measure of the difference between two probability distributions. For discrete distributions p, q with support $\{1, \dots, n\}$, the KL divergence is:

$$D(p||q) = \sum_{i=1}^n p(i) \log \frac{p(i)}{q(i)}$$

Show then, that $D(p||q) \geq 0$ for all p, q .

Hint: You may use this inequality without proof: $x - 1 \geq \log(x)$.